

Module 1 Assignment 3

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1.

```
## [1] NA
```

2.

```
## [1] 64.94737
```

3.

```
## [1] 63 60 65 NA 61 61 59 64 63 63 65 64 63 65
```

4.

```
## [1] "Mystery" "Hazel" "Spud" "Sandy"
```

5.

```
## [1] "Spud"
```

6.

```
## [1] "Spud"
```

8.

```
## group_id paddle_id length_in width_in depth_in spines insects damage location
## 1 1 1 12.5 8.5 0.50 Y Y Some Fourth
## 2 1 2 12.3 8.0 0.50 Y Y Some Fifth
## 3 1 3 14.5 11.0 0.70 Y Y Some Fifth
## 4 1 4 10.5 7.0 0.75 Y Y Some Fourth
## 5 1 5 8.5 6.5 0.20 Y Y Some Fourth
## 6 1 6 14.0 13.0 1.20 Y N Some Third
```

```
## temp_F species size
## 1 61 Opuntia engelmannii Medium
## 2 61 Opuntia engelmannii Medium
## 3 61 Opuntia engelmannii Medium
## 4 61 Opuntia engelmannii Medium
## 5 61 Opuntia engelmannii Medium
## 6 61 Opuntia engelmannii Medium
```

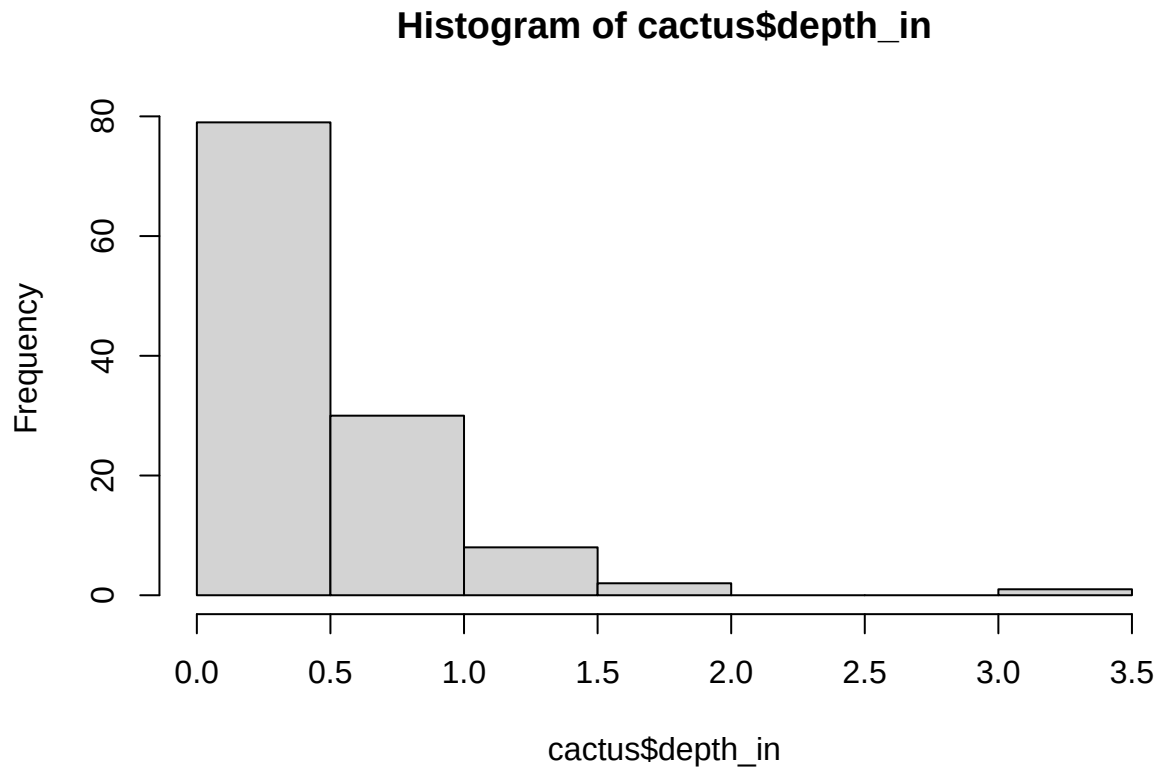
9.

```
## [1] 9.809542
```

10.

```
## [1] 9.8
```

11.



13.

```
## [1] 0.500 0.500 0.750 0.750 0.500 0.300 0.500 0.750 0.500 0.500 0.500 0.500
## [13] 0.250 0.500 0.500 0.500 0.500 0.250 0.400 3.200 0.500 0.750 1.750 0.500
## [25] 1.250 0.500 0.500 0.900 0.600 0.400 0.300 0.500 0.600 0.300 0.600 0.600
## [37] 0.400 0.600 0.500 0.600 0.500 0.500 0.500 0.300 0.400 0.250 0.500 0.250
## [49] 0.375 0.250 0.500 0.375 0.375 0.375 0.375 0.500 0.500 0.250 0.250 0.250
## [61] 0.500 0.250 0.250 0.250 0.250
```

14. Just a note that I am printing the first 10 rows of each data frame, not necessarily the entire thing, to include in the answer key.

```
##      group_id paddle_id length_in width_in depth_in spines insects damage
## 41         5         1      8.5     4.5     0.50      N       Y   Some
## 42         5         2      7.5     4.5     0.50      N       N   None
## 43         5         3     10.0     5.0     0.50      N       Y  Most
## 44         5         4     12.0     6.5     0.50      N       Y  Most
## 45         5         5      8.5     4.5     0.25      N       N   None
## 46         5         6      9.5     5.0     0.50      N       N   None
## 47         5         7     10.0     6.5     0.50      N       N   None
## 48         5         8      7.5     4.0     0.50      N       Y  Most
## 49         5         9      3.5     3.0     0.50      N       Y  Most
## 50         5        10      8.0     3.0     0.25      N       N   None
##      location temp_F      species  size
## 41    Third      61 Opuntia ficus-indica Large
## 42    Third      61 Opuntia ficus-indica Large
## 43 Fourth      61 Opuntia ficus-indica Large
## 44   Sixth      61 Opuntia ficus-indica Large
```

```
## 45    Fifth    61 Opuntia ficus-indica Large
## 46    Sixth    61 Opuntia ficus-indica Large
## 47    Sixth    61 Opuntia ficus-indica Large
## 48    Sixth    61 Opuntia ficus-indica Large
## 49  Seventh    61 Opuntia ficus-indica Large
## 50    Eighth    61 Opuntia ficus-indica Large
```

```
##      group_id paddle_id length_in width_in depth_in spines insects damage
## 1           1         1     12.5     8.5     0.50      Y      Y   Some
## 2           1         2     12.3     8.0     0.50      Y      Y   Some
## 3           1         3     14.5    11.0     0.70      Y      Y   Some
## 4           1         4     10.5     7.0     0.75      Y      Y   Some
## 5           1         5      8.5     6.5     0.20      Y      Y   Some
## 6           1         6     14.0    13.0     1.20      Y      N   Some
## 7           1         7     12.0    14.0     2.00      Y      Y   Some
## 8           1         8      9.0     7.5     1.00      Y      Y   Some
## 9           1         9     15.0    11.7     1.20      Y      Y   Some
## 10          1        10     12.8    10.0     0.90      Y      Y   Some
```

```
##      location temp_F      species      size
## 1     Fourth    61 Opuntia engelmannii Medium
## 2     Fifth    61 Opuntia engelmannii Medium
## 3     Fifth    61 Opuntia engelmannii Medium
## 4     Fourth    61 Opuntia engelmannii Medium
## 5     Fourth    61 Opuntia engelmannii Medium
## 6     Third    61 Opuntia engelmannii Medium
## 7     First    61 Opuntia engelmannii Medium
## 8     Third    61 Opuntia engelmannii Medium
## 9     Third    61 Opuntia engelmannii Medium
## 10    Fourth    61 Opuntia engelmannii Medium
```

15.

```
## [1] 5.242917
```

```
## [1] 8.341
```

16.

```
## [1] 1.782769
```

```
## [1] 2.256842
```